

Interoperability Solutions for Blockchain

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Abstract—In recent times, Blockchain has become one of the most influential technology. Numerous industry projects are developed and are under the development phase for enhancing blockchain technology in every way possible and making it applicable to various application-domains. Since the inception of Ethereum, many blockchain platforms have been developed using different types of technologies. Blockchain platforms, from its security perspective, don't interact with the outside world and acts as an independent and standalone ledger. But blockchain technology can not meet the needs of today's world if different ledgers don't interact with each other and with the legacy systems. Hence, Blockchain platforms should be able to communicate with each other and provide services to each other. Researchers and developers around the world are already working to achieve blockchain interoperability. In this paper, we survey interoperability solutions provided for blockchain platforms, both by industry people and the research community.

Keywords— *Blockchain Interoperability, Cross-Chain Technologies, Interledger Technologies, Inter-blockchain Communication.*

I. INTRODUCTION

Blockchain has gained tremendous popularity in recent years in industry and academia both, due to being an immutable and distributed ledger technology. Data in a blockchain is organized in the form of blocks linked to each other using hashes. Each peer in a blockchain network maintains a copy of the ledger, and its consistency across the peers is maintained using a consensus algorithm. The immutability and security of the distributed ledger are guaranteed by its consensus mechanism rather than any central organization. Many blockchain platforms have been developed since 2015 to achieve different types of functionalities. These functionalities are of two categories (i) improvements in scalability, privacy, and security of blockchain technology, (ii) enhancing its applicability to different domains. Blockchain platforms are mainly of two types: permission-less and permissioned. In a permission-less blockchain platform, anyone can join the network and use its functionality. However, in permissioned blockchains, users are authenticated peers of the network. These platforms work as standalone ledgers; they are not able to communicate with each other and use each other's services. This scenario is giving rise to the balkanization of blockchain. Also, there should not be a single ledger to rule all others. Blockchain platforms should be able to interact with each other to achieve a middle ground between the balkanization and maximization of the technology. Hence interoperability between blockchains is much required. Blockchain technology has the potential to facilitate decentralization of the World Wide Web and transform the world into a place created by people, governed by people and owned by people. However, its' full potential can only

be achieved if platforms are interoperable and can provide services to each other.

The focus of this paper is to survey interoperability techniques designed and developed for blockchain platforms. The rest of the paper is organized as follows: Section II provides a brief of blockchain interoperability and its use-cases. Section III presents an overview of different types of interoperability solutions. It also discusses the research work done related to different kinds of solutions. Sector IV discusses the advantages and disadvantages of different kinds of interoperability solutions. Section V concludes the paper with some direction for future work.

II. BLOCKCHAIN INTEROPERABILITY

Blockchain Interoperability is when one platform interacts with another to use its services. The interaction can be related to the referencing of data, verification of data, and use of the computational capacity of the blockchain network. Blockchain Interoperability can be achieved in many ways, as stated by [1]:

- Interaction between a legacy system and a Blockchain platform.
- Interaction between two Blockchain platforms.
- Interaction between smart-contracts of a decentralized application.

Interoperability among blockchains improves the scalability issue of blockchain technology. Off-chain payment channels and sharding techniques used for enhancing blockchain scalability implement interoperability concepts to some extent. Apart from improving scalability, blockchain interoperability is needed and is beneficial due to its other use-cases. The literature review has revealed the following use-cases: -

- **Asset Exchange:** Assets are exchanged simultaneously on different ledgers. User A sends assets on one blockchain to the user to B and, in return, receives assets on another blockchain from user B. Assets remain on their respective blockchains and are just transferred from one account to another.
- **Asset Transfer:** Assets are transferred from one blockchain platform to another. For transfer, assets on both blockchains should be compatible with each other. While transferring, assets are destroyed or burnt on one blockchain and are created on another blockchain. So, that total supply of the asset remains the same.
- **Cross-Chain Oracles:** A smart contract on a blockchain platform reads data from another platform and performs actions accordingly.

III. INTEROPERABILITY SOLUTIONS

A. Notary Scheme Solutions

Notary schemes use a centralized architecture by using a trusted third party for carrying out asset exchange/transfer. A trusted third party employed by notary schemes can be a single notary [2] or a consortium of notaries implementing a multi-signature wallet [3]. The latter scenario is known as a decentralized notary scheme.

Polkadot [4] is a protocol to carry out transactions between blockchain networks smoothly. Polkadot uses the concept of the Relay chain and Parachains. The Relay chain is used for registering and finalizing transactions between Parachains. Parachain can be any permissioned or permission-less blockchain. Relay chains use the DOT token to carry out all the management activities using proof of stake consensus algorithm.

Cosmos [5] is a project to create an internet of blockchains where heterogeneous blockchains can communicate with each other. Cosmos have a Tendermint core, a BFT based consensus engine. Tendermint core interacts with the application layer using The Application Blockchain Interface (ABCI). Cosmos network consists of a Cosmos hub and zones that are Tendermint based Blockchains. Zones interact with Cosmos hub using Inter-Blockchain Communication (IBC) protocol. Different zones can interact with each other via the Cosmos hub. A special type of zone called peg-zone is used to interact with blockchains, which are not Tendermint based like Ethereum.

Ripple [6] is a digital payment network for global money transfer. It was initially launched in 2012. Ripple has its own distributed ledger known as Ripple consensus ledger. XRP is ripple's cryptocurrency. Transactions on the Ripple network are cheaper than on the Bitcoin or Ethereum network. Users can make payments across all types of ledgers from fiat currency to cryptocurrencies on blockchains. It is mostly used for cross-border payments.

ICON [7] projects aim to create a decentralized network where different blockchains under different governance can communicate with each other. The crypto token of the project is ICX. ICON project consists of a nexus blockchain based on Loop Fault Tolerance (LFT) consensus algorithm, which is an improvement of the BFT algorithm. Nexus is connected to other community blockchains through ICON republic portals. This project is exclusively designed for Korea and follows the regulations and policies for blockchain and cryptocurrencies as allowed by Korea.

Metronome [8] is a project that hopes to provide autonomous cryptocurrency. The token used by this project is MET, and as of now, this token can hop between Ethereum and Ethereum classic. The aim of the project is to allow its token to chainhop between different blockchains. Implementation of chainhops for QTUM and RSK is under the testing phase.

Interchain [9] is a proposed framework for carrying out communication among blockchains. It consists of interchain, subchain, interchain nodes, validating nodes, and gateway nodes. A transaction is carried out between two subchains and is managed by validating nodes and gateways nodes. Three-way handshaking is used for the asset transfer.

The proposed framework is theoretical and doesn't provide any information regarding the consensus algorithm used.

B. Sidechain/ Relay Solutions

A sidechain is a blockchain connected to the main blockchain with a two-way peg. Two-way peg connectivity is used to transfer assets between sidechain and mainchain. A sidechain can work independently on its own and can have implementation different from the parent chain. A sidechain is primarily used for improving the scalability of the main blockchain or for adding additional functionality to it. Transaction delays are usually less on sidechains. Two-way peg transfer can be designed using the following three approaches [10]: -

Centralized two-way peg: A trusted third party is used for transferring assets between sidechain and mainchain. Locking and unlocking of assets are solely handled by a trusted third party. It's very easy to implement and provides extremely fast transfer of assets. This type of implementation of sidechain suffers from a single point of failure. Hence, it promotes centralization against the core idea of the decentralization of blockchain technology.

Federated two-way peg: In this approach, instead of a single party controlling the asset transfer, a group of entities or notaries have the control of locking and unlocking of assets. Transfer transactions happen if the majority of notaries sign the transaction. Unlike the above-mentioned centralized approach, this approach doesn't suffer from a single point of failure and promotes decentralization. Federated two-way peg can suffer from an internet attack and a majority of entities lose their private keys.

Two-way based on Simplified Payment Verification (SPV) proofs: SPV proofs are used to guarantee that a transaction is added in a legitimate block of the blockchain upon which new blocks have been added to form the longest chain. This type of two-way peg approach is slow due to the confirmation time taken by the transactions in a blockchain but is decentralized in nature.

Elements [11] project designed the first successful federated pegs for bitcoin blockchain in 2016. It's built on bitcoin codebase, so it acts as a testbed for bitcoin improvements also. Elements platform is an open source project and capable of operating as a standalone blockchain as well as a sidechain pegged to another blockchain. It provides the feature of asset issuance, confidential transactions, and signed blocks.

Liquid [12] is a sidechain developed by the Blockstream and is based on the Elements framework. It's based on a strong federation consisting of cryptocurrency exchanges and various financial institutes. It supports the issuance of assets where an organization can issue a tokenized version of its assets. These assets can then be traded using the Liquid platform. It provides permissioned blockchain with faster speed for trading and better privacy than bitcoin.

Rootstock(RSK) [13] is sidechain pegged to bitcoin blockchain via a two-way peg. It is a smart contract execution platform hence provides additional functionality to the bitcoin blockchain. RSK implements the concept of merged mining, where it can benefit from the miners of the bitcoin blockchain to provide mining on the RSK platform. It provides Bitcoin with a Turing complete RootStock Virtual Machine (RVM) for executing smart contracts

parallelly. RVM can execute Solidity smart contracts as it is compatible with Ethereum Virtual Machine (EVM). RSK's coin is Smart Bitcoins (SBTC). It improves the scalability of the Bitcoin blockchain and can be used for retail payments or in supply chain traceability.

Plasma [14] is a sidechain developed for Ethereum blockchain. Each sidechain is implemented by creating a smart contract on Ethereum main chain. Each sidechain has its rules and constraints which are imposed by its smart contract. Plasma chains implement the Proof-of-Stake consensus algorithm, but mining is done on the Ethereum mainchain. Plasma chains manage the production of blocks by customizing the permissioned set of nodes. Transactions and state changes happening are stored only on sidechains. Node validators just provide the periodic activity information in the form of blockheader hashes to the Ethereum chain. Plasma chains provide scalability to decentralization applications on Ethereum. Plasma is used by projects like OmiseGO [15], whose intent is to provide faster peer-to-peer financial transactions across the globe.

LOOM [16] project is also based on the concept of plasma chains. It provides an SDK for creating and deploying Decentralized Applications (DApps) on their sidechains, which are pegged to Ethereum using a two-way peg. Loom uses delegated proof of stake algorithm to provide consensus among its validators. Loom transactions are not settled immediately on Ethereum mainchain but are settled in bulk. Its focus is to provide a highly scalable and usable DApps building platform to developers.

POA [17] is an open-source Ethereum sidechain network for developing smart contract applications on sidechains. Its consensus is Proof-of-Authority. It is implemented as a two-way peg, and all its validators are United States based notaries. POA supports native solidity smart contracts, thus making smart contracts portable between Ethereum and POA network. The main focus of the POA project is to provide scalability and interoperability between Ethereum and sidechains.

C. Smart Contract Solutions

A transaction between two parties is considered atomic when a transaction happens entirely or not at all. Atomic cross-chain trade is when two users using different blockchains want to exchange assets without any trusted third-party. Trades like this are also known as atomic swaps. Atomic swaps record transactions on both the blockchains involved in trading using hash-locks [18] and time-locks [19] without involving any third-party. Hash-lock and time-locks bound a recipient to admit the payment of a transaction by generating a cryptographic proof within a certain time-limit. Thus, atomic swaps use Hashed Time-Lock Contracts (HTLCs) [20], which is a smart contract to avoid counterparty risks. Atomic cross-chain transactions are also used by other interoperability solutions like bridging approaches discussed further.

Borkowski *et al.* [21] proposed a protocol for transferring the same type of token on the same kind of blockchains (Ethereum) in a multiple blockchain environment. Robinson *et al.* [22] proposed a methodology for processing atomic transactions using smart contract function calls across consortium sidechains. Bennink *et al.* [23] proposed single-use smart contracts for the exchange

of coins and tokens between Ethereum blockchain. Maurice Herlihy [24] has modeled cross-chain swaps as directed graphs and provided a protocol for atomic swaps. Li *et al.* [25] proposed the architecture for satellite chains, which are independent chains of a signal blockchain. These chains can run in parallel to achieve scalability and can transfer assets to each other. Dagher *et al.* [26] proposed a patient record management system using public and private Ethereum Blockchains along with a local database.

D. Bridging Solutions

These interoperability solutions provide a bridge between two chains to transfer/exchange assets. Bridging solutions are different from notary schemes and sidechains as they don't involve any main chain. Smart contracts or other modules are used as a bridge to connect two blockchains.

Blocknet [27] is an open-source project launched in 2014 to provide a decentralized network for different blockchains to connect and communicate. It also enables blockchains to connect to off-chain APIs and oracle services. Blocknet's utility token is BLOCK, and it is Proof-of-Stake based service blockchain. It consists of three main components: - XRouter, XBridge, and XCloud. XRouter provides a communication layer to Blocknet protocol to allow applications to interact with blockchains and verify data without downloading the full blockchain. XBridge provides a trustless exchange of digital assets supported by Blocknet with the help of APIs. XCloud allows applications to use off-chain data on-chain.

ARK [28] is also an open-source project, and its mainnet was launched in May 2017. It uses the concept of SmartBridge to connect different blockchains. ARK ecosystem uses a Delegated Proof-of-Stake consensus algorithm and acts as a bridge between interconnecting blockchains. New blockchains can be easily built using ARK core technology and can then connect to ARK mainnet. Two types of protocols are used for SmartBridge communication: - Protocol-Specific SmartBridge to connect ARK based blockchains and Protocol-Agnostic SmartBridge to connect ARK based blockchain to other types of blockchains.

Wanchain blockchain [29] is a public blockchain forked from Ethereum. It supports cross-chain functionality and ring signature-based private transactions. It implements Proof-of-stake as its consensus algorithm. A user can create smart contracts, build DApps, and issue token on it. Wanchain's main aim is to provide cross-chain transmission of assets across any type of blockchain. But as of now, it has been integrated with Bitcoin, Ethereum, EOS, and some ERC-tokens. Token traded on Wanchain is WAN.

Aion Project [30] aims to implement a multi-tier blockchain network for different blockchains to interoperate. It follows a hub-spoke model, where interconnected chains interact with the hub at the center through spokes. Aion network consists of four main components: Connecting Networks, Participating Networks, Interchain transactions, and Bridges. Connecting networks are the protocols to connect different participating blockchains with the Aion ecosystem. Participating networks can have their own consensus algorithms and governance. Bridges implement a BFT-

based consensus algorithm to perform interchain transactions. Aion implements handshaking and Proof-of-intelligence as a consensus algorithm.

BTC Relay [31] is a smart contract deployed on Ethereum blockchain that reads and verifies bitcoin transactions. BTC Relay stores the bitcoin block headers and creates a mini version of Bitcoin blockchain on Ethereum. Relayers submit the block headers to BTC relay and get a fee when a transaction in a block is verified. BTC Relay is the first Bitcoin-Ethereum cross-chain implementation released in 2016. Users can pay in bitcoins to use Ethereum smart contracts through BTC Relay.

E. Blockchain Router Solutions

The studies [32][33][34] use the blockchain node as a router to achieve communication between blockchain platforms. Node architecture of the node acting as a router needs to be changed to achieve interoperability among blockchains. All the above-mentioned studies in this category are proposed solutions and do not take in to count the problem of the finality of a transaction into consideration.

Lightening network [35] and Raiden [36] are scalability solutions for Bitcoin and Ethereum respectively. These solutions support off-chain micropayment channels to carry out transfer of value between two users. Only the initial transaction which initiates the channel between two parties and the final transaction which closes the channel after the transfer of value is recorded on blockchain. Lightening network and Raiden are also considered as interoperable solutions by some studies in the literature. Table I shows some of the studies carried out in literature related to interoperability solutions for blockchain.

TABLE I. SUMMARY OF SURVEYS RELATED TO BLOCKCHAIN INTEROPERABILITY

Studies	Year	Observations
Kannengiesser <i>et al.</i> [37]	2020	Properties and characteristics of a cross-chain technology (CCT), patterns and use-cases of CCT, reports 57 artifacts, of CCT
Siris <i>et al.</i> [38]	2019	Survey of all categories of interoperability solutions and their comparison
T. Koens <i>et al.</i> [1]	2019	Describes a set of key properties for interoperable solutions and compare Polkadot and Cosmos projects.
Singh <i>et al.</i> [10]	2019	An extensive review of the sidechains along with their advantages and disadvantages.
Qasse <i>et al.</i> [39]	2019	Survey of four types of interoperability solutions (Sidechains, blockchain routers, Smart contract, Industry Solutions)
Johnson <i>et al.</i> [40]	2019	Describes some of the sidechain solutions (Plasma, Clearmatics, Metronome, Polkadot, TAST)

Zamyatin <i>et al.</i> [41]	2019	Formally defines the problem of Correct Cross-Chain Communication and reduction into a fair exchange problem
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IV. DISCUSSION

Most of the interoperability solutions like sidechains, notary scheme, bridging solutions discussed in this study use intermediaries to achieve cross-chain communication. Current interoperability solutions mostly connect permission-less ledgers or permissioned ledgers. Further research is required to connect permission-less ledgers and permissioned ledgers, as the latter doesn't share information with other systems, and the privacy of information is crucial for such ledgers. In an interoperable solution, the state finality between deterministic ledgers and probabilistic ledger is a major issue and requires further investigation.

Some of the sidechain solutions are not open-source; hence the empirical analysis of such sidechains is difficult and costly for researchers. Hash locks and time locks are the simplest interoperability solutions, while sidechains and notary schemes are complex but are highly scalable. Several interoperability articles are proposals only and lack practical implementation. Table II shows an overview of the limitations of interoperability solutions discussed in this study.

TABLE II. SHORTCOMINGS OF INTEROPERABILITY SOLUTIONS

Interoperability Solution type	Shortcomings
Notary Schemes	Single point of failure, highly complex.
	The state exchanged between two ledgers is stored at three places.
	Works on the assumption that most of the participants are honest. It tends towards centralization due to the presence of TTP.
Sidechain/ Relays	Implements transfer of assets in a 1 to 1 relationship between sidechain and mainchain.
	It focuses only on homogeneous blockchain systems.
	Federated two-way peg suffers from political centralization. Higher computational cost and highly complex.
Smart Contract	Smart contract space is explored less for implementing the interoperable solutions. Smart contract sharing is not available
Bridging solutions	Only a few bridging solutions implement asset exchange; others implement asset transfer only. Higher computational cost
Blockchain Router	Proposed solutions are not implemented yet. The node structure of the platform needs to be changed so it can act as a router.

V. CONCLUSION AND FUTURE WORK

This Survey presents the interoperability solutions implemented for blockchain by industry and academia both. The focus of this study was to identify different categories of solutions proposed or implemented to achieve an interoperable ecosystem of blockchains. There are only a few smart contract solutions for achieving interoperability;

thus, it needs more investigation by researchers. Most of the sidechain's solutions discussed in the literature are based on federated two-way peg architecture, and there is a lack of research on SPV based two-way peg-based sidechains.

Blockchain interoperability is still in its infancy stage. There are no established standards for connecting blockchains. Hence a lot of research is required and is under its way as well to make blockchains fully interoperable and communicate with each other without using any trusted third party.

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